

SpaceX Falcon 9 First Stage Landing Prediction

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29 July 2026

<https://github.com/EliRoux/Applied-data-science-Capstone>

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
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- Discussion
 - Findings & Implications
- Conclusion
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EXECUTIVE SUMMARY

Project Objective

- The objective of this project was to develop a machine learning model capable of predicting whether the first stage of a SpaceX Falcon 9 rocket would land successfully. Accurate predictions of landing success can help estimate launch costs and assess mission risk, as reusable boosters significantly reduce the overall cost of space launches.

Key Findings

- Falcon 9 landing success has improved steadily over time.
- Launch site, orbit type, payload mass, and booster version influence landing success.
- Machine learning models were able to predict landing outcomes with high accuracy.
- Interactive visualisations provided valuable insights into launch trends and mission performance.

Data Collection

Data Wrangling

Exploratory Data Analysis

Interactive Visualisation

Machine Learning

Landing Success Prediction



INTRODUCTION

Background

- SpaceX has transformed the aerospace industry by developing reusable launch vehicles. Unlike traditional rockets, the Falcon 9 is designed to return its first-stage booster safely after launch, allowing it to be refurbished and reused for future missions. This innovation has significantly reduced the cost of space travel.

Business Problem

- The cost of a Falcon 9 launch depends heavily on whether the first-stage booster can be recovered successfully. Predicting landing success is valuable because it helps estimate mission costs, evaluate operational risk, and support launch planning.

INTRODUCTION

Project Objective

- The primary objective of this project is to build a machine learning model that predicts whether a Falcon 9 first-stage booster will land successfully using historical launch data collected from the SpaceX API and Wikipedia.

Project Questions

- Which launch characteristics influence landing success?
- How do payload mass, orbit type, and launch site affect mission outcomes?
- Which machine learning algorithm provides the most accurate predictions?



Data Collection Methodology

Data Sources

SpaceX REST API

- Retrieved structured launch data in JSON format.
- Collected information such as:
 - Flight Number
 - Launch Date
 - Rocket ID
 - Payload ID
 - Launch Site
 - Booster Version
 - Orbit
 - Landing Outcome

Wikipedia

- Used web scraping with BeautifulSoup to extract launch records not directly available through the API.
- Retrieved additional launch information including:
 - Payload Mass
 - Mission Outcome
 - Booster Details
 - Launch History

Tools and Libraries

- Python
- Requests
- Pandas
- BeautifulSoup

Data Wrangling Methodology

Data Cleaning Process

- After collecting the data, several preprocessing steps were performed to prepare it for analysis and machine learning.

Cleaning Steps

- Converted JSON responses into Pandas DataFrames.
- Merged data collected from the API and Wikipedia.
- Removed Falcon 1 launches, retaining only Falcon 9 launches.
- Identified and handled missing values.
- Replaced missing Payload Mass values using the mean payload mass.
- Standardised column names and data formats.
- Created a binary target variable (Class) representing landing success:

Feature Engineering

- Additional features were prepared for modelling, including:
 - One-hot encoding of categorical variables.
 - Standardisation of numerical features.
 - Creation of training and testing datasets.



Exploratory Data Analysis (Visualisation Results)

- **Key Findings**
- **Landing success improved over time**, indicating continual improvements in SpaceX's landing technology.
- **Payload mass influences landing success**, with certain payload ranges showing higher success rates.
- **Orbit type affects landing outcomes**, with some orbits achieving consistently higher success rates than others.
- **Launch site performance varies**, reflecting differences in mission profiles and operational experience.
- **Conclusion**
- Exploratory data analysis identified several variables—particularly launch site, payload mass, orbit type, and flight number—as important predictors of landing success. These insights informed the feature selection process for the machine learning models.



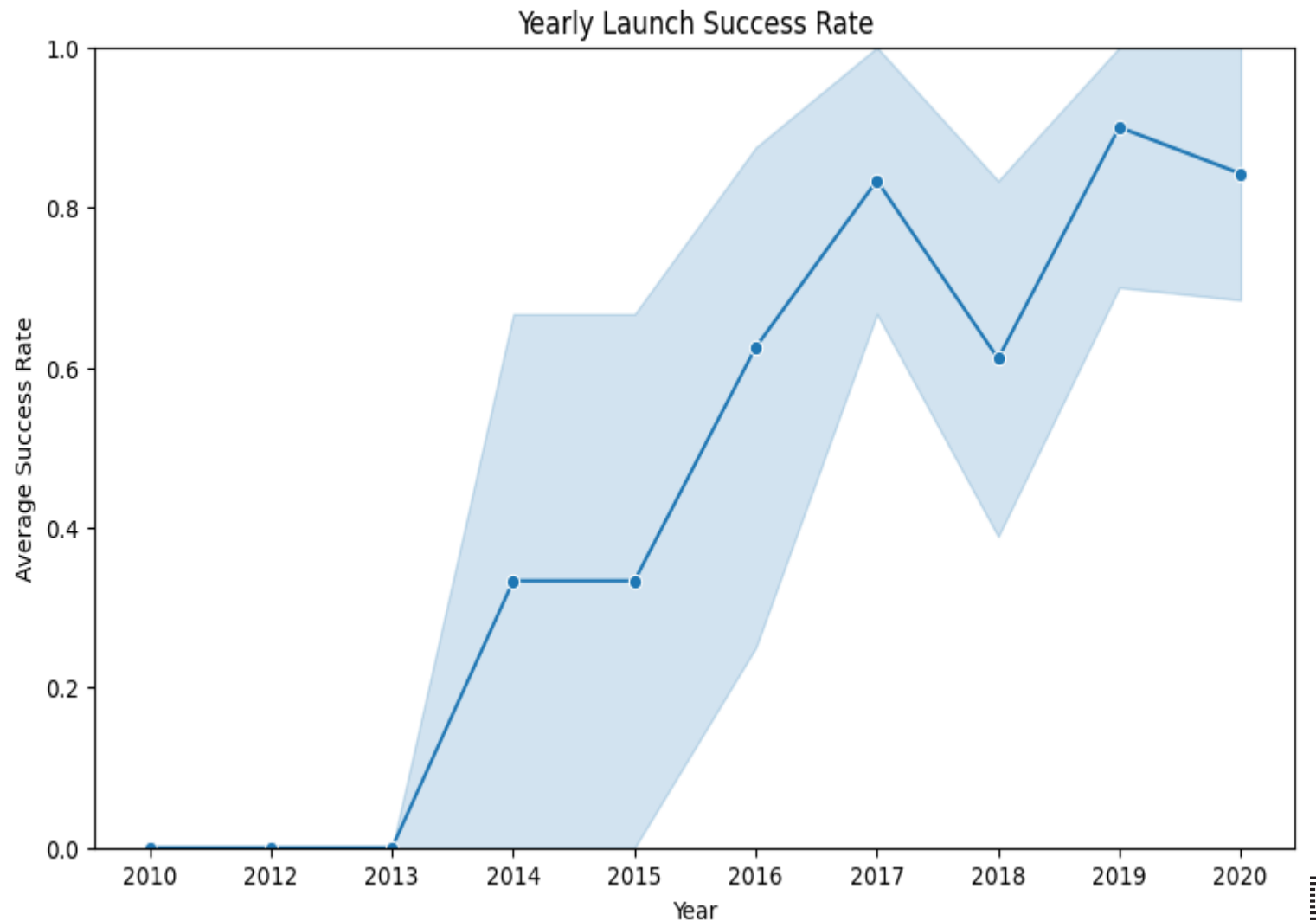
Exploratory Data Analysis (Visualisation Results)

- **Observation**

Between 2014 and 2020, the average success rate increased significantly, rising from around **35% in 2014** to approximately **90% in 2019**, before remaining high at around **85% in 2020**.

- **Key Insight**

The increasing success rate over time indicates that **flight year is an important predictor of landing success** and reflects the maturation of SpaceX's reusable launch system.



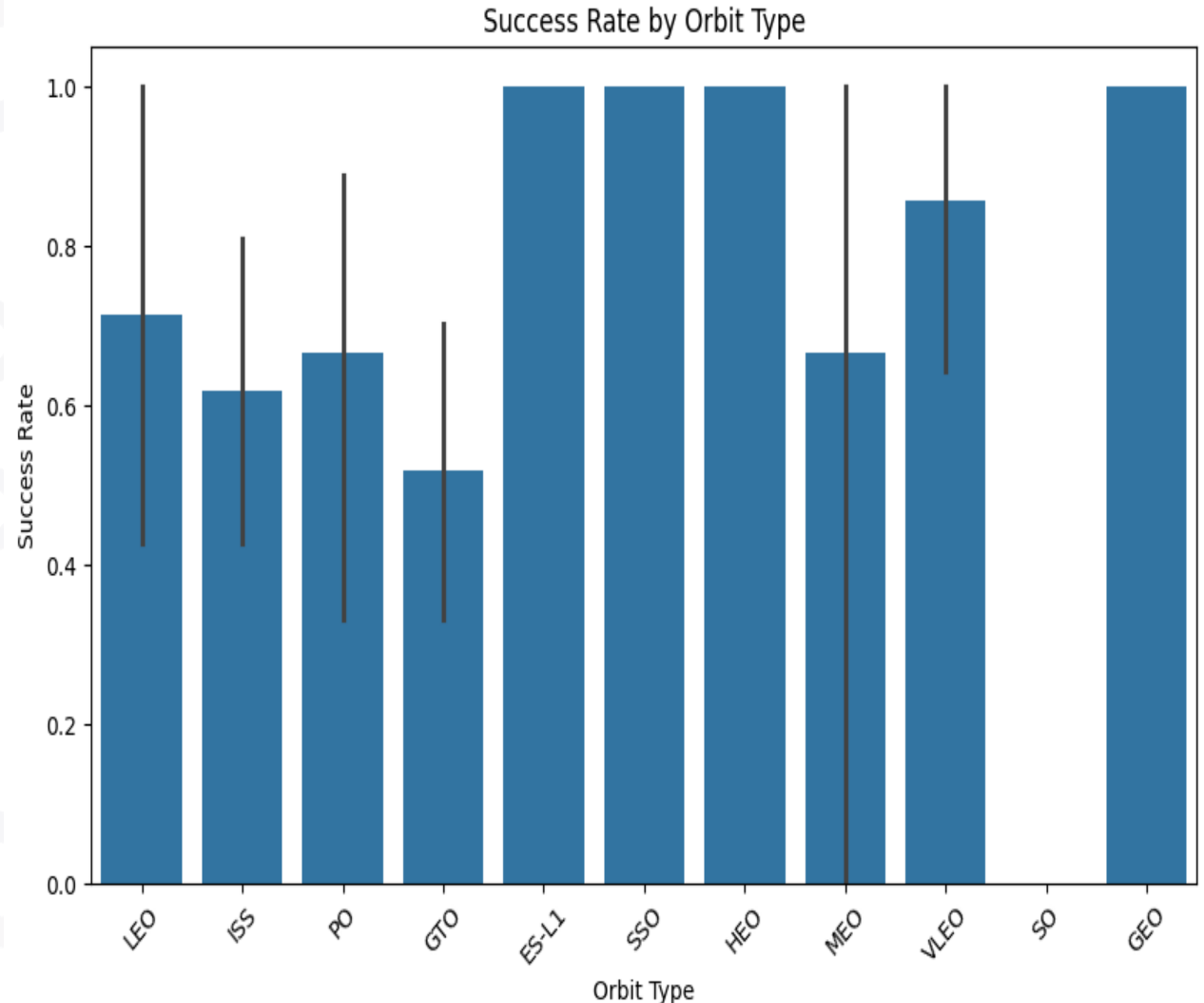
Exploratory Data Analysis (Visualisation Results)

- **Observation**

The bar chart compares the average landing success rate across different orbit types. The **ES-L1, SSO, HEO, and GEO** orbits achieved a **100% landing success rate**, indicating that all recorded Falcon 9 launches to these orbits resulted in successful first-stage landings. In contrast, **GTO (Geostationary Transfer Orbit)** had the lowest success rate at approximately **50%**.

- **Key Insight**

The results suggest that **orbit type is an important factor influencing Falcon 9 landing success**. However, orbit types with very few observations should be interpreted cautiously, as their 100% success rates may not be representative of future missions due to the limited sample size.



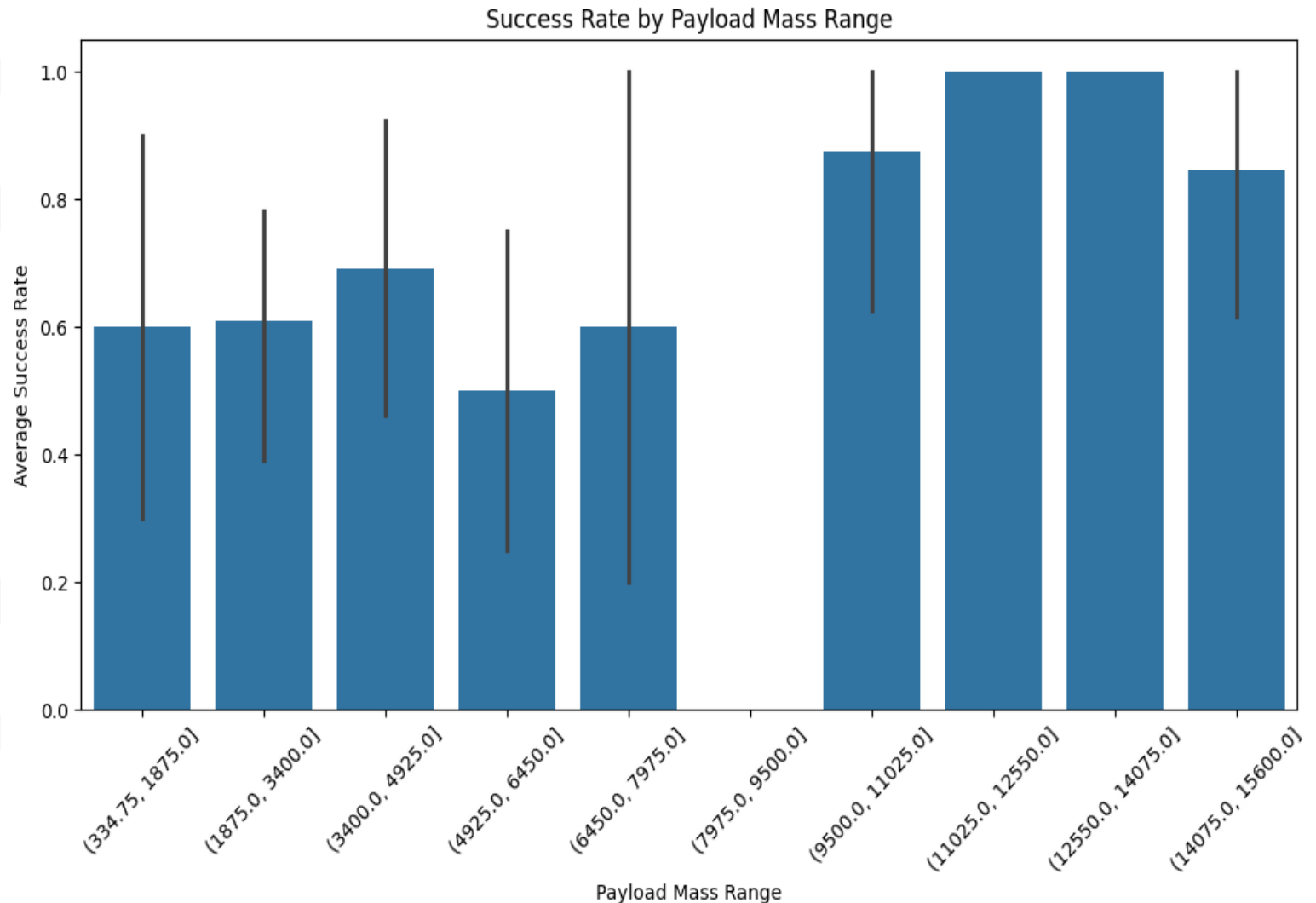
Exploratory Data Analysis (Visualisation Results)

- **Observation**

This chart groups payload masses into ranges (bins) and compares the average landing success rate for each range. The results show a general upward trend, with **higher payload mass ranges achieving higher average landing success rates**.

- **Key Insight**

The analysis indicates that **payload mass is associated with landing success**, although this relationship should not be interpreted in isolation. Improvements in booster technology, operational experience, and launch procedures over time likely contributed to the higher success rates observed for larger payloads.



SQL Analysis Results

- **Key Findings**

- Falcon 9 launches increased steadily over the years.
- Successful landings became more common as booster technology improved.
- Certain booster versions consistently carried heavier payloads.
- Launch sites supported different mission types and payload profiles.

- **Conclusion**

- SQL analysis complemented the visual EDA by validating trends and enabling efficient exploration of structured launch data.



Interactive Visual Analytics

- **Key Findings**

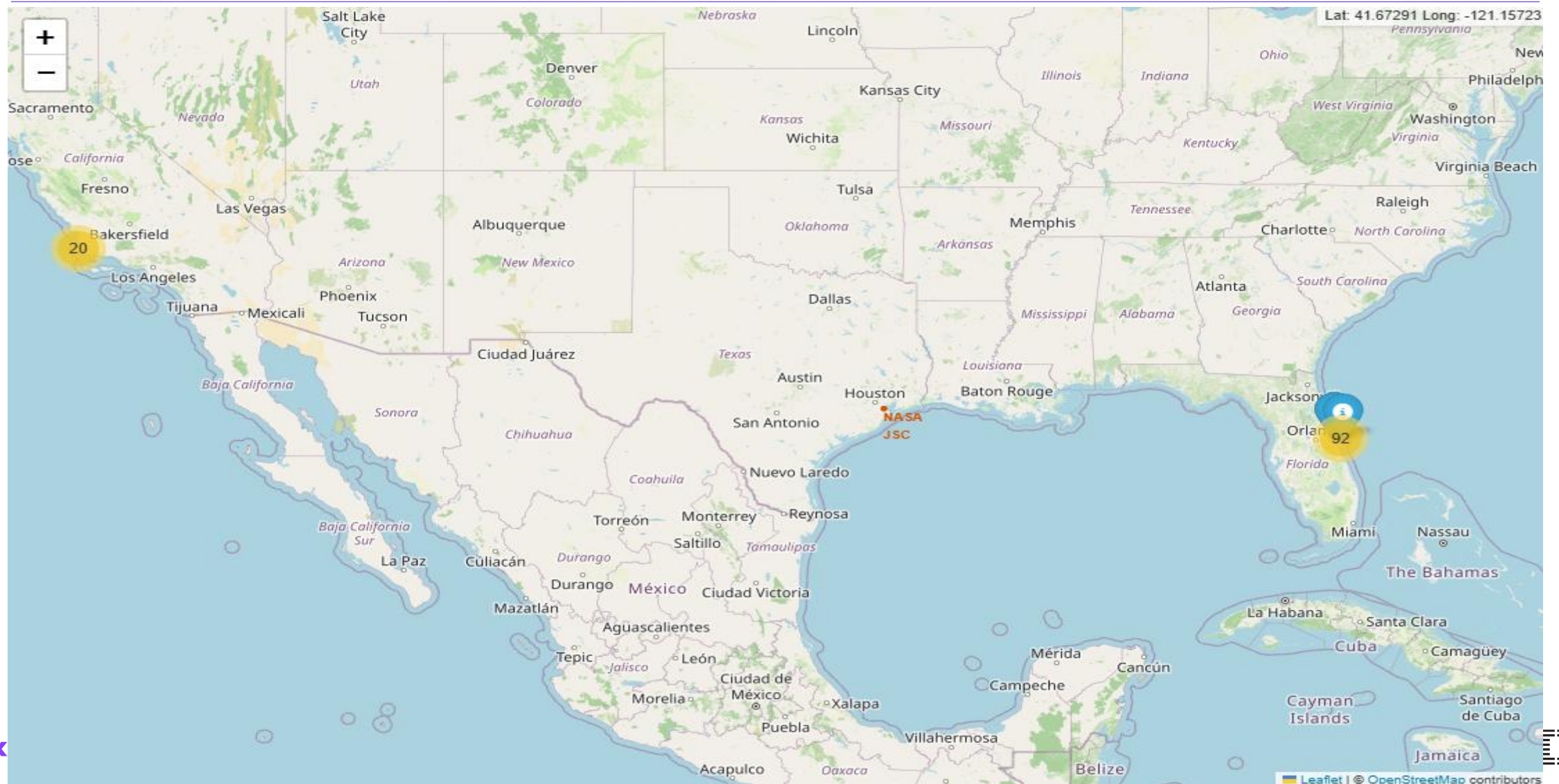
- Interactive maps provide geographical context for launch operations and illustrate why launch sites are located near coastlines.
- The dashboard enables users to dynamically explore how launch site selection and payload range influence landing success.
- Colour-coding by booster version highlights differences in performance across Falcon 9 variants.

- **Conclusion**

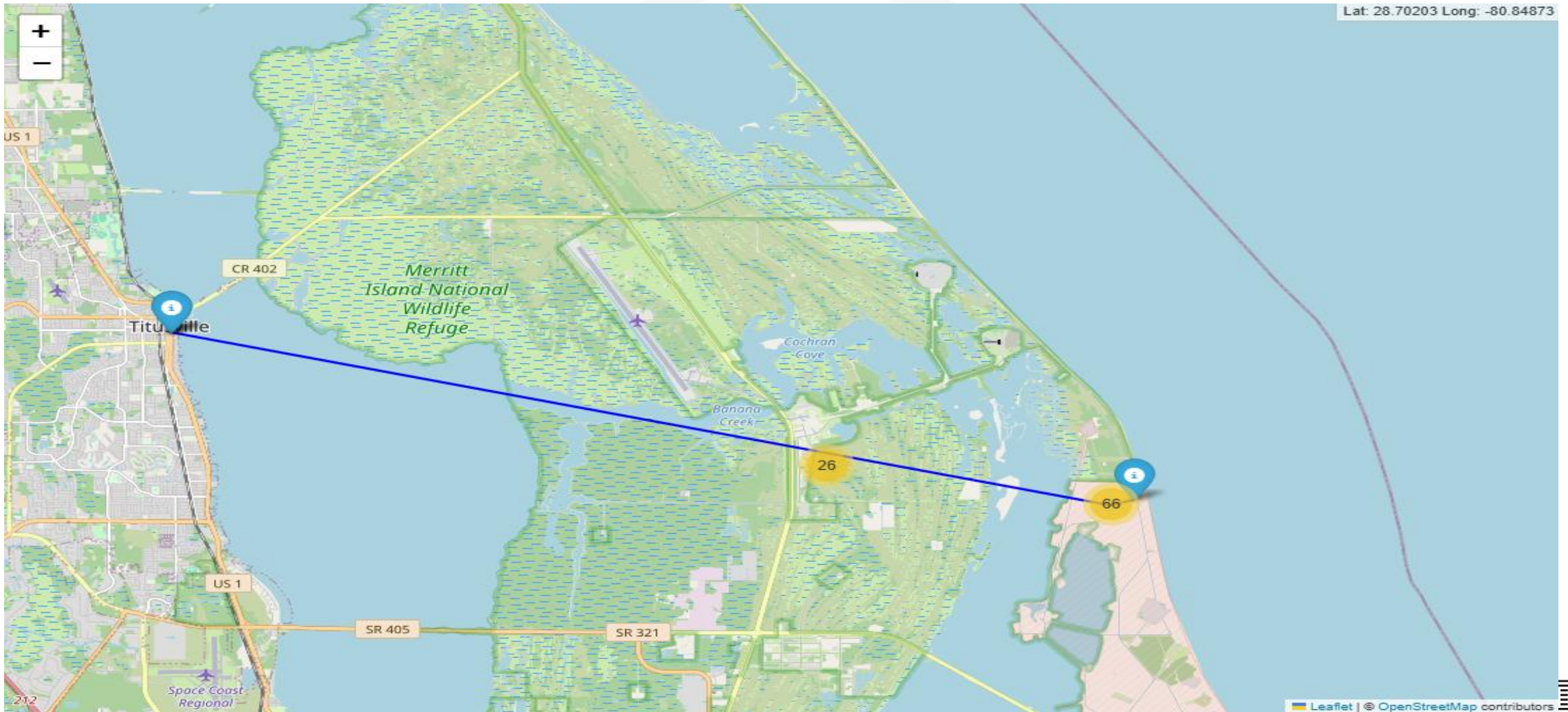
- Interactive visualisation tools make complex launch data easier to explore and reveal relationships that may not be immediately apparent in static charts.



Interactive map screenshot 1



Interactive map screenshot 2



Are launch sites in close proximity to railways?

- **Answer:** Yes.
- **Explanation:**

The launch sites are generally located relatively close to railway infrastructure. Railways provide an efficient means of transporting large rocket components, fuel, and other equipment required for launch operations. Having rail access reduces transportation costs and simplifies logistics for moving oversized cargo.

Are launch sites in close proximity to highways?

- **Answer:** Yes.
- **Explanation:**
All of the launch sites are situated near major highways, allowing personnel, equipment, and support vehicles to travel efficiently to and from the launch facilities. Easy road access is essential for daily operations, maintenance activities, and emergency response.

Are launch sites in close proximity to the coastline?

- **Answer:** Yes.
- **Explanation:**

The launch sites are intentionally built close to the coastline. Launching rockets over the ocean minimizes the risk to populated areas in the event of a launch failure or the shedding of rocket stages. Coastal locations also provide safe flight paths and simplify recovery operations for reusable Falcon 9 boosters landing on drone ships or returning to nearby landing zones.

Do launch sites keep a certain distance away from cities?

- **Answer:** Yes.
- **Explanation:**

The launch sites are located a considerable distance from major cities. This separation helps protect the public from noise, vibrations, and potential hazards associated with rocket launches. It also provides a safety buffer in case of launch anomalies while still allowing reasonable access for employees and support infrastructure.

DISCUSSION



Key Findings

- The **Falcon 9 landing success rate improved significantly over time**, increasing from approximately **50% in 2014** to around **90% in 2019**, demonstrating the continuous improvement of SpaceX's reusable rocket technology.
- **Orbit type** influenced landing success, with **ES-L1, SSO, HEO, and GEO** achieving the highest observed success rates, while **GTO** missions had the lowest success rate due to their greater energy requirements.
- A positive relationship was observed between **payload mass** and landing success, although this trend is likely influenced by improvements in booster technology and operational experience over time.
- Interactive **Folium maps** showed that launch sites are strategically located near coastlines and transport infrastructure while remaining at a safe distance from major cities.
- The **Plotly Dash dashboard** enabled interactive exploration of launch success by launch site and payload range, making it easier to identify patterns in the data.

DISCUSSION



Predictive Analysis

- Several machine learning classification algorithms were evaluated using hyperparameter tuning and cross-validation. The models demonstrated that historical launch data can be used to predict first-stage landing success with high accuracy, confirming that variables such as launch site, orbit type, payload mass, and booster version are valuable predictors.



DISCUSSION



Limitations

- Some orbit categories contained relatively few launches, meaning their observed success rates may not be representative.
- External factors such as weather conditions, technical issues, and operational decisions were not included in the dataset and may also affect landing outcomes.
- The analysis was based on historical launch data up to 2020 and does not account for more recent missions.

DISCUSSION



Overall Discussion

- The results demonstrate that **historical launch characteristics provide valuable insights into Falcon 9 landing success.** Combining exploratory data analysis with machine learning enabled both the identification of important influencing factors and the development of predictive models capable of estimating landing outcomes. This illustrates how data science techniques can support decision-making in aerospace operations.

OVERALL FINDINGS & IMPLICATIONS

- **Overall Findings**

- Falcon 9 landing success rates improved significantly over time, reflecting advancements in SpaceX's reusable rocket technology.
- Launch success was influenced by several key factors, including **orbit type, payload mass, launch site, and booster version**.
- Interactive visualisations and SQL analysis revealed meaningful patterns and relationships within the launch data.
- Machine learning models successfully predicted landing outcomes with high accuracy using historical launch data.

- **Implications**

- Predictive models can support **mission planning** and **risk assessment** by estimating the likelihood of successful booster recovery.
- The findings demonstrate how data science can be used to improve **operational decision-making** in the aerospace industry.
- Successful booster recovery contributes to **lower launch costs**, making space missions more economically sustainable.
- The project highlights the value of combining **data collection, exploratory analysis, visualisation, and machine learning** to solve real-world problems.



CONCLUSION



- **Project Summary**
- This project applied the complete data science methodology to analyse and predict the landing success of SpaceX Falcon 9 first-stage boosters. Data was collected from the SpaceX API and Wikipedia, cleaned and prepared for analysis, explored using visualisations and SQL, and used to train and evaluate multiple machine learning classification models.

CONCLUSION



Key Conclusions

- Falcon 9 landing success has improved significantly over time, demonstrating the effectiveness of SpaceX's reusable rocket programme.
- **Launch site, orbit type, payload mass, and booster version** were identified as important factors influencing landing success.
- Exploratory data analysis and interactive visualisations provided valuable insights into launch performance and operational trends.
- Machine learning models successfully predicted landing outcomes, showing that historical launch data can be used to forecast future mission success.

CONCLUSION



Future Work

- Incorporate additional features such as **weather conditions, wind speed, and sea state** to improve prediction accuracy.
- Expand the dataset with more recent Falcon 9 launches and future missions.
- Investigate advanced machine learning techniques, such as **XGBoost, Random Forests**, or **Neural Networks**, to compare predictive performance.
- Deploy the predictive model as a web application to support real-time mission planning and decision-making.

APPENDIX



- **Tools and Technologies**
- **Programming Language:** Python
- **Data Collection:** SpaceX REST API, Wikipedia Web Scraping
- **Data Processing:** Pandas, NumPy
- **Data Visualisation:** Matplotlib, Seaborn, Plotly Express
- **Interactive Mapping:** Folium
- **Dashboard Development:** Plotly Dash
- **Machine Learning:** Scikit-learn
- **Database Queries:** SQLite
- **Development Environment:** Jupyter Notebook
- sqlite3
- **Data Sources**
- **SpaceX REST API:** <https://api.spacexdata.com>
- **Wikipedia Falcon 9 Launch Records:** https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



APPENDIX

Machine Learning Models Evaluated

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

Key Python Libraries

- pandas
- numpy
- matplotlib
- seaborn
- plotly
- folium
- scikit-learn
- requests
- BeautifulSoup4
- sqlite3



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